Perimeter & Area

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Easy / Standard

1. 16cm

2. 16cm

3. 12cm²

4. C — both A (20cm) and B (20cm)

5. 20cm

6. 3cm

7. C — 16cm²

8. 3cm

9. A, C and E (perimeters 14cm, 10cm and 16cm are all under 18cm)

10. 20cm

11. 9cm²

12. B — 4×4=16cm² is larger than 5×3=15cm²

13. 24cm

14. A square; 25cm²

15. 4cm

Standard

16. 22cm

17. C — both A (22cm) and B (22cm) have a perimeter of 22cm

18. 30cm²

19. 20cm

20. 5cm

21. B — 27cm²

22. 48m

23. A, B and D (areas 24, 25 and 21 cm² are all more than 20cm²)

24. 40cm²

25. A — 8×3=24cm², matching the 6×4=24cm² rectangle

26. 30cm

27. 81cm²

28. 18cm²

29. 90cm

30. 4cm

Standard / Hard

31. 60 − 4 = 56 m² (the patio's area is 2×2 = 4m²)

32. A — 36 − 6 = 30cm²

33. 30cm (the rectangle formed is 10cm by 5cm, so perimeter = 2×(10+5))

34. 22cm (length = 4+3 = 7cm; perimeter = 2×(7+4))

35. Area = 5×4 = 20m²; cost = 20 × £8 = £160

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36. Side = 6m (since 6×6=36); perimeter = 4×6 = 24m

37. A, B and D (perimeter 30cm, area 50cm², and it has 4 right angles)

38. 200 − 12 = 188cm²

39. 60 − 8 = 52m²

40. C — 2×(50+30) = 160 m

41. 36 ÷ 4 = 9cm

42. C — a 5cm×9cm rectangle has the largest perimeter (28cm; the others are 24cm, 24cm and 26cm)

43. 12 + 4 = 16m²

44. New dimensions are 10cm×6cm, so the area = 60cm² (4 times the original 15cm²)

45. Outer side = 10 + 1 + 1 = 12m; perimeter = 4 × 12 = 48m

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Homework Worksheet — Answers

Easy / Standard

1. 18cm
2. 20cm
3. 14cm²
4. C — both A (16cm) and B (16cm)
5. 24cm
6. 4cm
7. C — 25cm²
8. 4cm
9. A and C (perimeters 16cm and 12cm are both less than 20cm; B=24cm and D=26cm are more, and E=20cm exactly, not less)
10. 20cm
11. 10cm²
12. B — 4×4=16cm² is larger than A's 6×2=12cm²
13. 28cm
14. A square; 36cm²
15. 5cm

Standard

16. 26cm
17. C — both A (24cm) and B (24cm)
18. 42cm²
19. 24cm
20. 6cm
21. C — 40cm²
22. 60m
23. A and C (areas 30cm² and 28cm² are more than 24cm²)
24. 45cm²
25. A — 3×10=30cm², matching the 5×6=30cm² rectangle
26. 32cm
27. 49cm²
28. 16cm²
29. 74cm
30. 4cm

Standard / Advanced

31. Length = 2×4 = 8cm; perimeter = 2×(8+4) = 24cm
32. A — larger area = 4×9 = 36cm², so side = 6cm (since 6×6=36)
33. The rectangle formed is 10cm by 5cm; perimeter = 2×(10+5) = 30cm
34. Rectangle's perimeter = 2×(9+3) = 24cm; square's side = 24 ÷ 4 = 6cm
35. 36 − 1 = 35m²
36. 36cm (the rectangle formed is 12cm by 6cm)
37. Area = 4×3 = 12m²; cost = 12 × £7 = £84

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38. Side = 8m (since 8×8=64); perimeter = 4×8 = 32m

39. A — 40 − 4 = 36m²

40. A and B (perimeter 28cm, area 40cm²; the other statements are false)

41. 150 − 9 = 141cm²

42. B — 2×(40+20) = 120 m

43. 64 ÷ 8 = 8cm

44. 12 + 4 = 16m²

45. Outer area = 14×8 = 112m²; decking = 112 − 72 = 40m²

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